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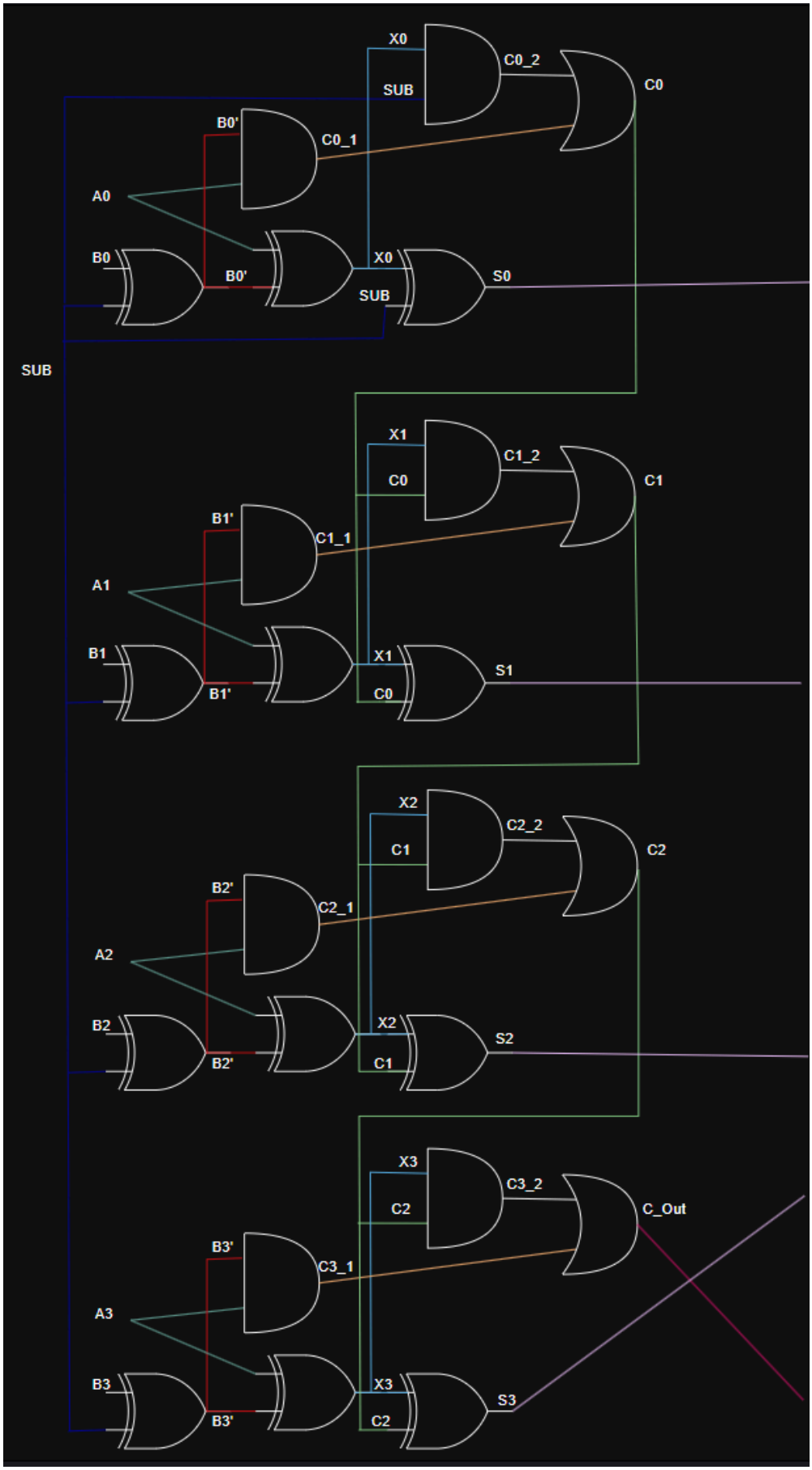
# Bonus Project 1 - 4 Bit Addition / Subtraction

The purpose of this report is to design, implement, and simulate a 4-bit adder/subtractor using only basic logic gates. The circuit performs both binary addition and subtraction by using XOR, AND, and OR gates instead of Full-adder chips. The structure of this report will include the logic flow diagram, circuit design, and the simulation process.

## Part 1: Logic Flow Diagram

This diagram aims to visualize the entire summation and subtraction process, bit by bit. The diagram was created using the Draw.io Website.

The subtraction control line is connected to an XOR gate at each B input. When Sub = 0, the XOR gates allow the original B values to pass unchanged, enabling normal binary addition. When Sub = 1, the XOR gates invert the B inputs, producing the one's complement of B. The same SUB signal is also used as the initial carry-in for the first stage. At each stage, the first XOR gate modifies the B input according to the SUB value. The second XOR gate combines the B value with the corresponding A input to generate an intermediate signal. The final XOR gate combines this value with the carry-in from the previous stage to produce the sum output.

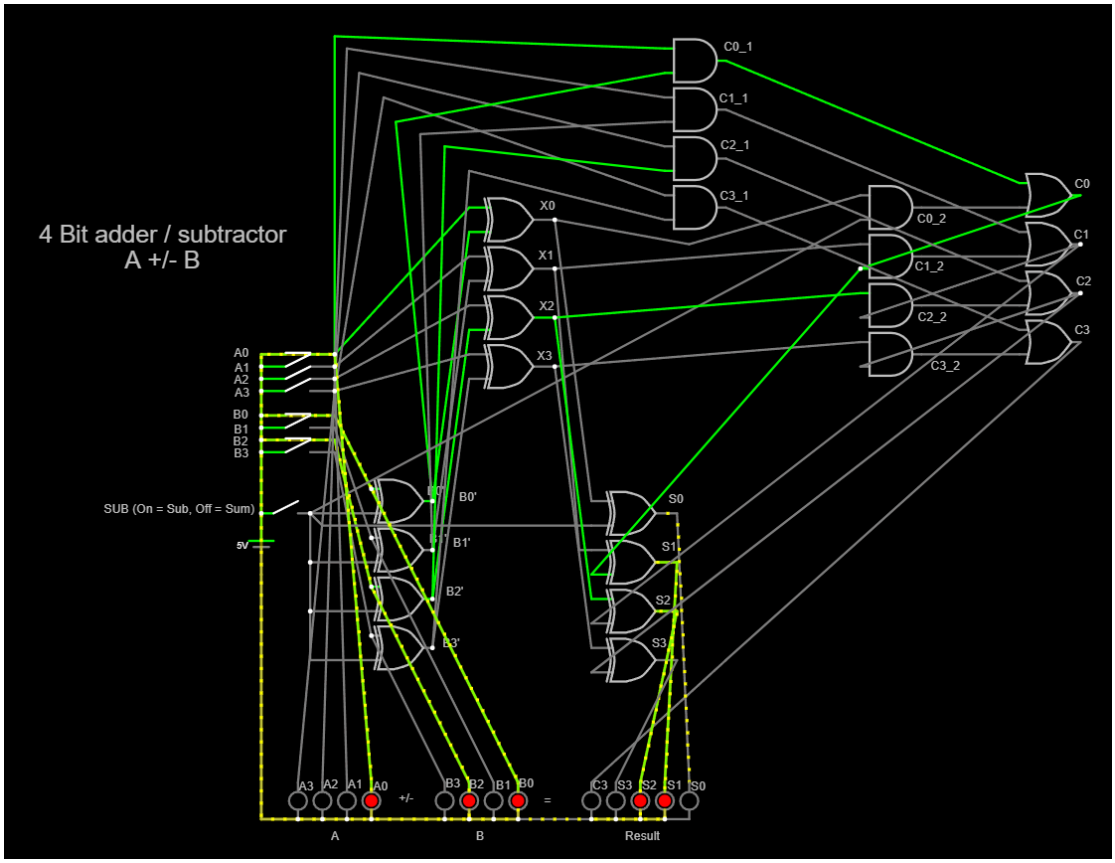

X0
C0_2
C0
SUB
B0'
C0_1
A0
B0
B0'
X0
S0
SUB
SUB
X1
C1_2
C1
C0
B1'
C1_1
A1
B1
X1
S1
B1'
C0
X2
C2_2
C2
C1
B2'
C2_1
A2
B2
X2
S2
B2'
C1
X3
C3_2
C_Out
C2
B3'
C3_1
A3
B3
X3
S3
B3'
C2

Carry generation is achieved through two AND gates and one OR gate. The first AND gate generates a carry from the A and modified B inputs (Cx_1), while the second AND gate generates a carry from the intermediate XOR result and the previous carry (Cx_2). These carry terms are combined using an OR gate to produce the carry-out for the stage (Cx).

The carry-out from each stage is used in the next stage. The final outputs are represented by S0 through S3, while the final carry-out is represented by C_out

Part 2: Online Simulation

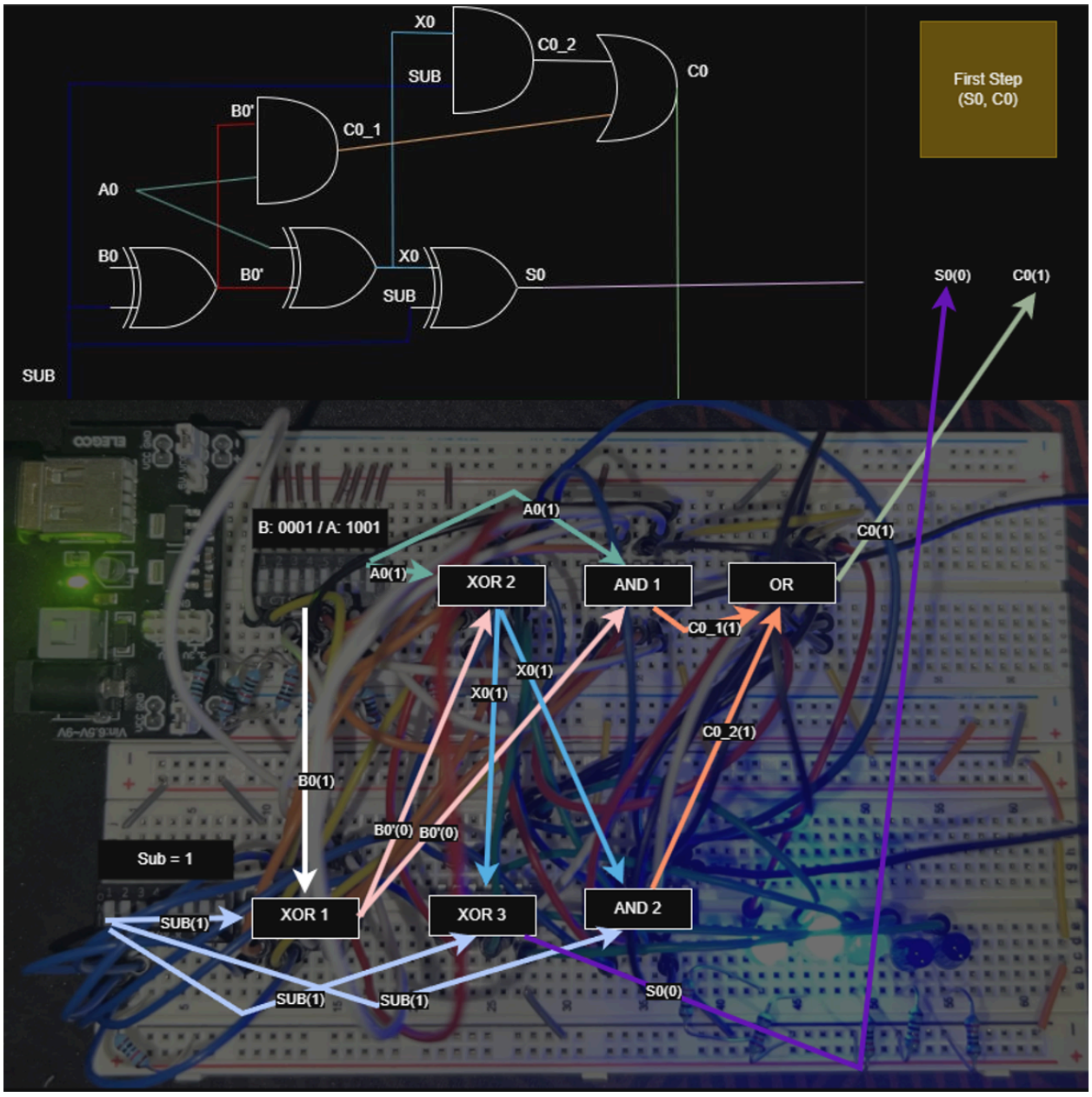
Here is the link to the interactive circuit as represented above, made in the Falstad circuit simulator, with an image of how it looks.



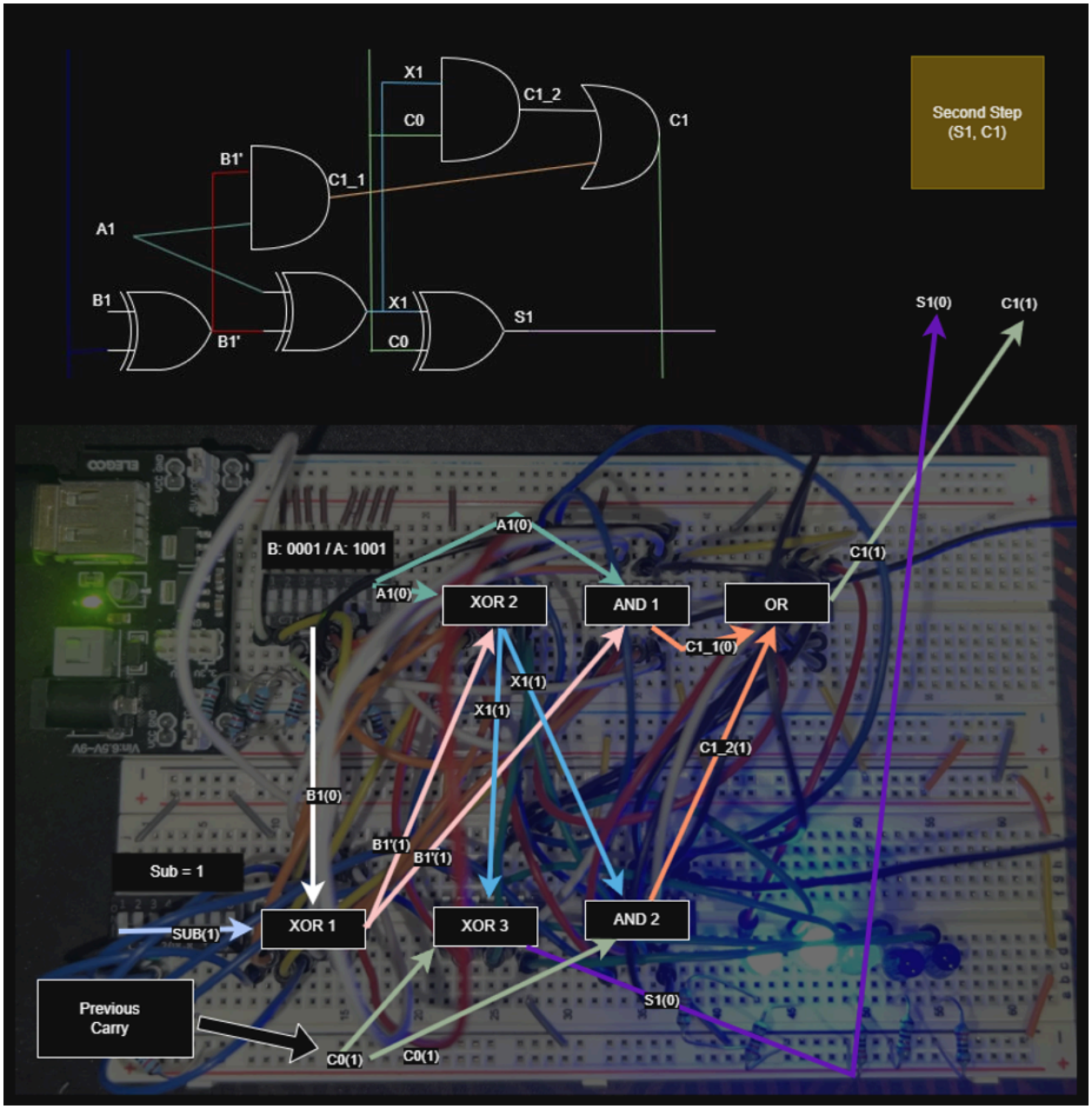
(https://tinyurl.com/468p2ah8)

Part 3: Step-by-Step analysis of the example A(1001) - B(0001)

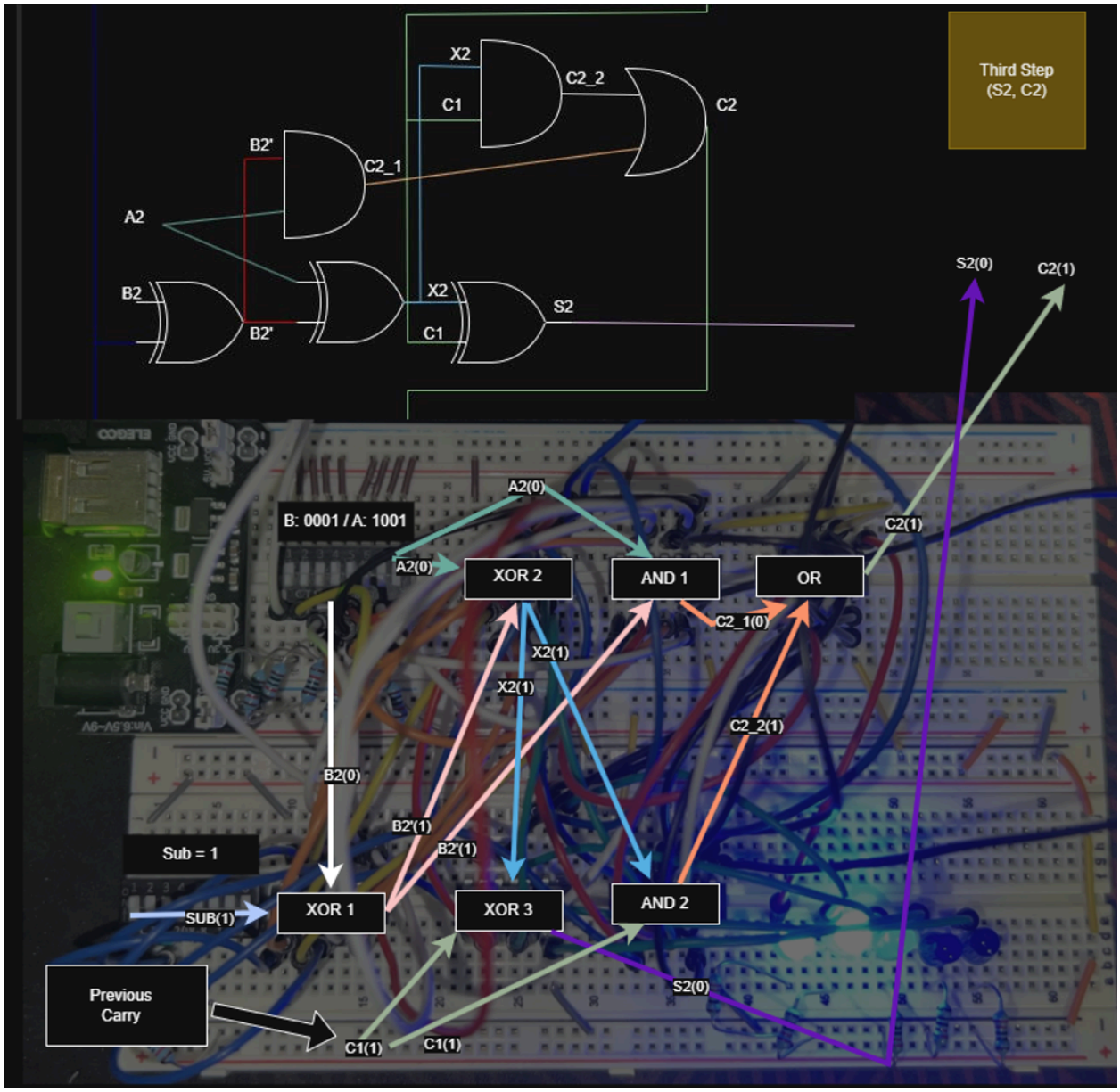
Step 1 (S0, C0):



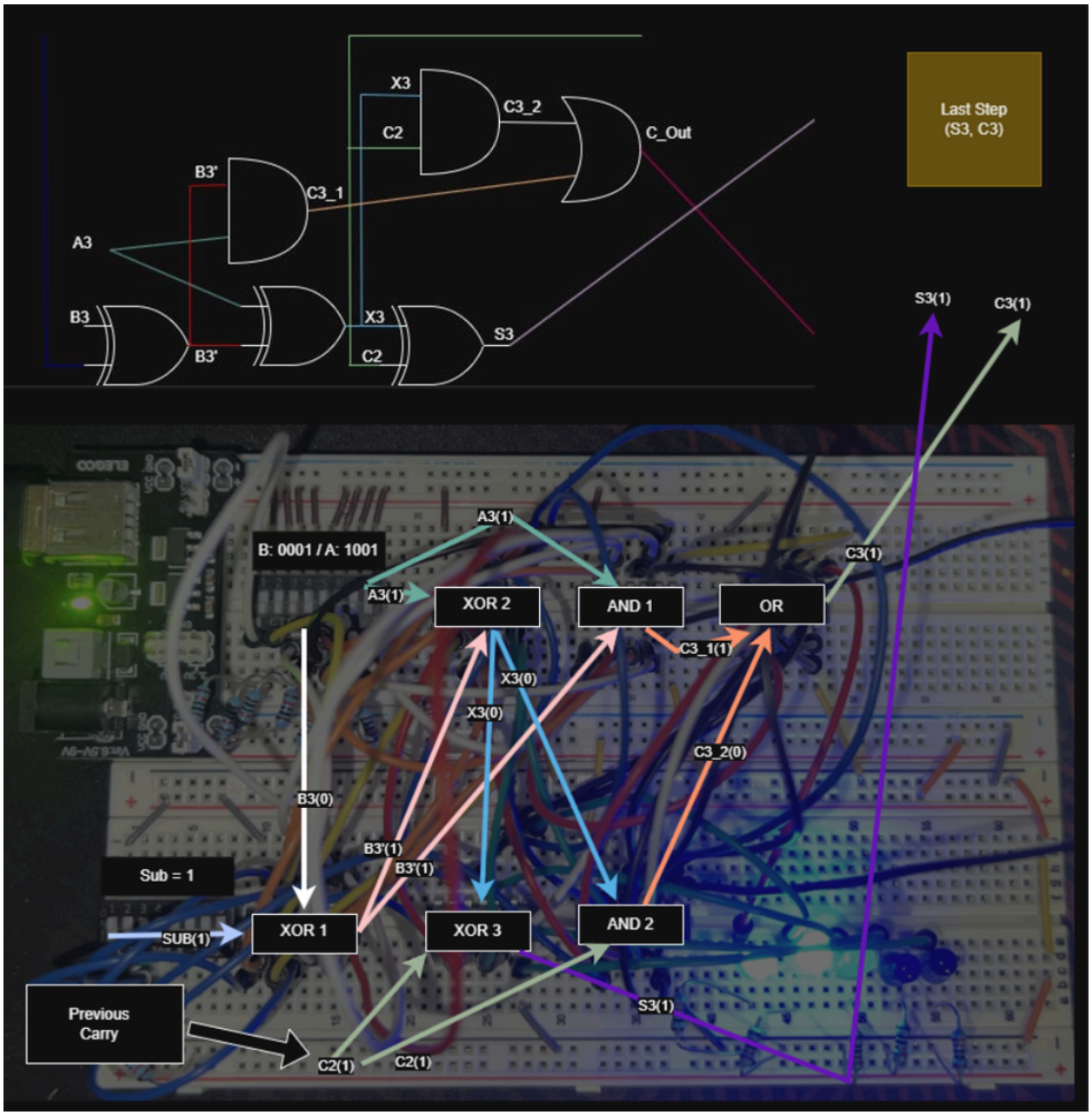
Step 2 (S1, C1):


Second Step
(S1, C1)
X1
C0
C1_2
C1
B1'
C1_1
A1
B1
X1
C0
S1
B1'
S1(0)
C1(1)
B: 0001 / A: 1001
A1(0)
A1(0)
XOR 2
AND 1
OR
C1(1)
C1_1(0)
X1(1)
X1(1)
C1_2(1)
B1(0)
B1'(1)
B1'(1)
Sub = 1
SUB(1)
XOR 1
XOR 3
AND 2
Previous
Carry
S1(0)
C0(1)
C0(1)

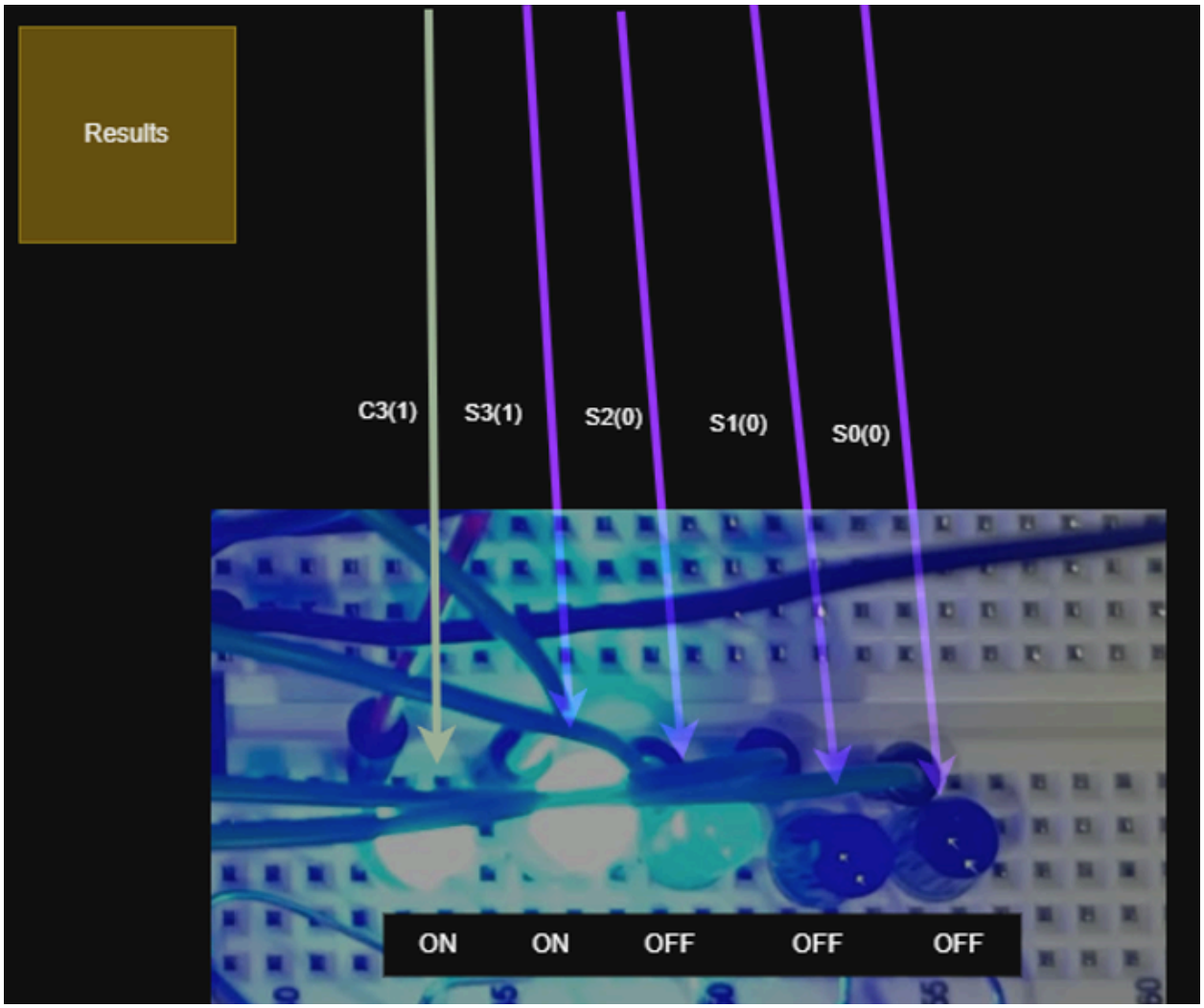
Step 3 (S2, C2):


Third Step
(S2, C2)
X2
C2_2
C1
C2
B2'
C2_1
A2
B2
B2'
X2
C1
S2
S2(0)
C2(1)
B: 0001 / A: 1001
A2(0)
A2(0)
XOR 2
AND 1
OR
C2(1)
C2_1(0)
X2(1)
X2(1)
C2_2(1)
B2(0)
B2'(1)
B2'(1)
Sub = 1
SUB(1)
XOR 1
XOR 3
AND 2
S2(0)
Previous
Carry
C1(1)
C1(1)

Step 4 (S3, C3):


X3
C3_2
C2
C_Out
B3'
C3_1
A3
B3
X3
S3
B3'
C2
Last Step
(S3, C3)
S3(1)
C3(1)
B: 0001 / A: 1001
A3(1)
A3(1)
XOR 2
AND 1
OR
C3(1)
C3_1(1)
X3(0)
X3(0)
C3_2(0)
B3(0)
B3'(1)
B3'(1)
Sub = 1
SUB(1)
XOR 1
XOR 3
AND 2
Previous
Carry
S3(1)
C2(1)
C2(1)

Results:




The final result produced by the circuit for the operation A(1001) − B(0001) is (1000). The output matches and is mathematically correct, as 1001 = 9, 0001 = 1. 9 - 1 = 8. And 1000 = 8.

## Conclusion

This project successfully demonstrates the implementation of a 4-bit adder/subtractor using only XOR, AND, and OR logic gates. Both the Falstad simulation and physical breadboard implementation verified the correctness of the arithmetic operations and carry logic.